



Use the Built-in Security Features in Your FOSS Distro

When it comes to securing network infrastructure, the trend is to invest in commercial-grade appliances. Sadly, security designers tend to ignore the built-in security features available in FOSS distros. This article explains a few such features, and will help readers decide when and how to use them in a given network scenario.

Before we talk about the various features built into FOSS systems, we need to understand some terminology as well as the various attack methods. A network is typically divided into LAN and WAN, wherein the LAN can have network switches to further sub-divide it into manageable small networks. LAN and WAN are separated by a router. A firewall is deployed within the network to keep a watch on network packets from the security standpoint. This division of the internal and external network is called a perimeter, and as mentioned earlier, the firewall is an important component of the perimeter. Viewing firewall operations at a high level, the policies of what should be allowed to get into the network from the outside world, and what on the Internet should be accessible from the internal network decide the 'perimeter defence'. In a complex network scenario, perimeter defence can also incorporate intrusion detection systems, UTM (Unified Threat Management) devices, etc, to make it more robust.

In the cyber security world, most attacks happen from within the network—perhaps due to a root-kit, spy-ware, or a virus making its way to desktops or servers; or by a

disgruntled employee who knows enough about the internal network to plant an attack. How an attack takes place is also important to understand. Please refer to the table below, which classifies various security attacks into two categories—network and Web.

Network attacks	Web attacks
Denial of Service	Denial of Service
Network sniffing	Cross-site scripting
Packet crafting	Man in the middle
Packet spoofing	Cryptographic attacks
Device invasion	Database hacking
DNS attacks	Cookie injection
Wireless attacks	Code exploitation

As you can see, this is an elaborate list of attacks, and no single device can stop them all and protect the network. For example, since a firewall sits on the perimeter, it hardly helps to prevent internal attacks, prompting a need to have an application-level firewall. Similarly, an anti-virus running on servers may not be capable of performing packet-level security analysis, which is crucial for a mission-critical database server. Cyber security is never 100 per cent fool-